COMPUTER NETWORKING

Networking Fundamentals

How devices connect, communicate and stay secure



What We Will Cover

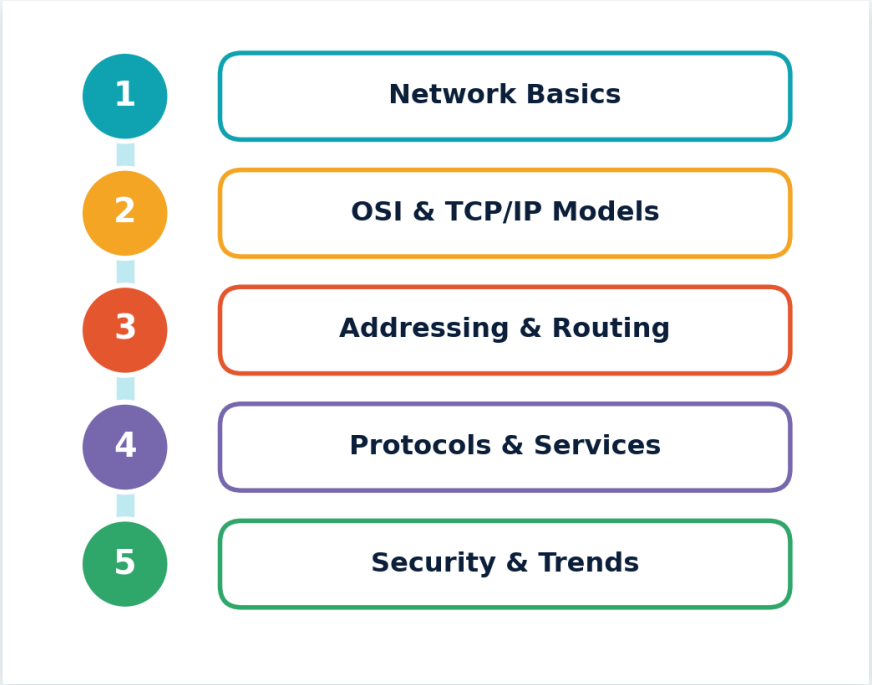
1 **Network basics** devices, types, topologies

2 **Models** OSI and TCP/IP

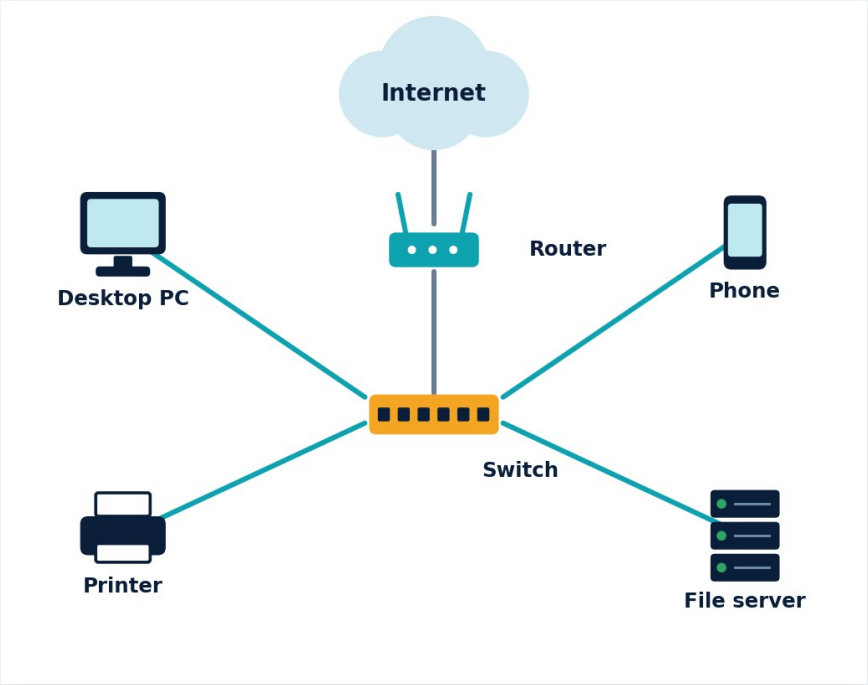
3 **Addressing and routing** IP, subnets, paths

4 **Protocols** TCP, UDP, DNS, DHCP, Wi-Fi

5 **Security and trends** threats, defenses, cloud



What Is a Network?



Definition two or more devices linked to share data

Link cables or wireless signals

Shares files, printers, apps and internet access

Scale from 2 computers to the whole Internet

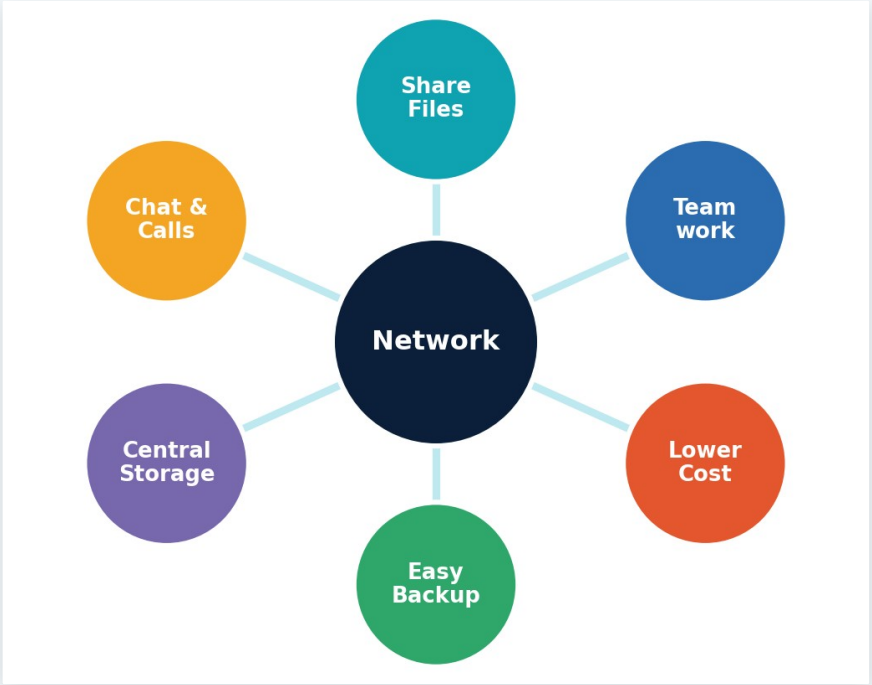
Why Networking Matters

Sharing printers, files and one internet line

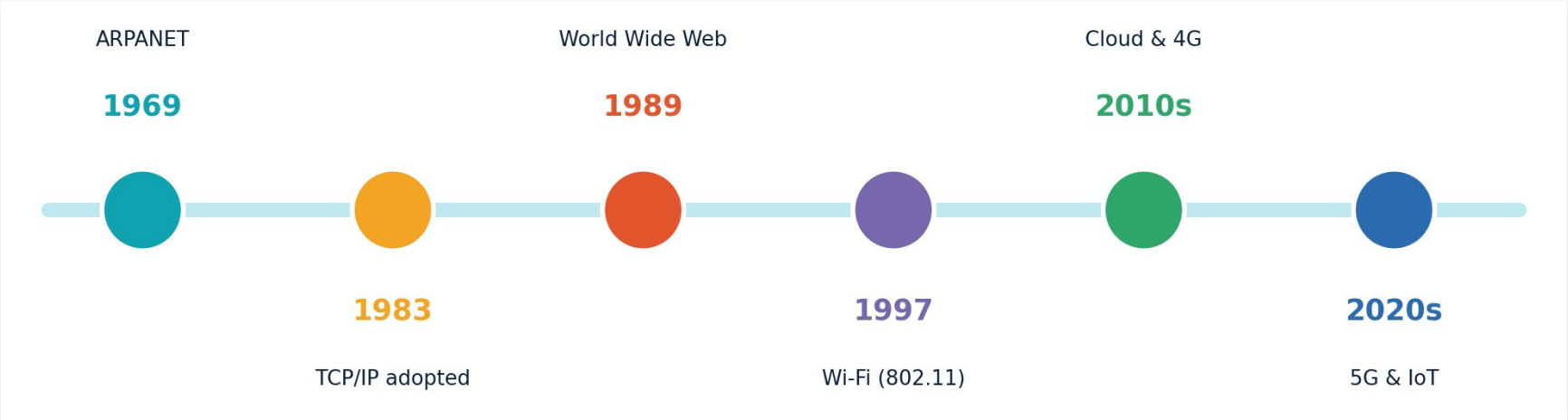
Communication email, chat and video calls

Central storage and simple backups

Savings less hardware, better teamwork



A Short History



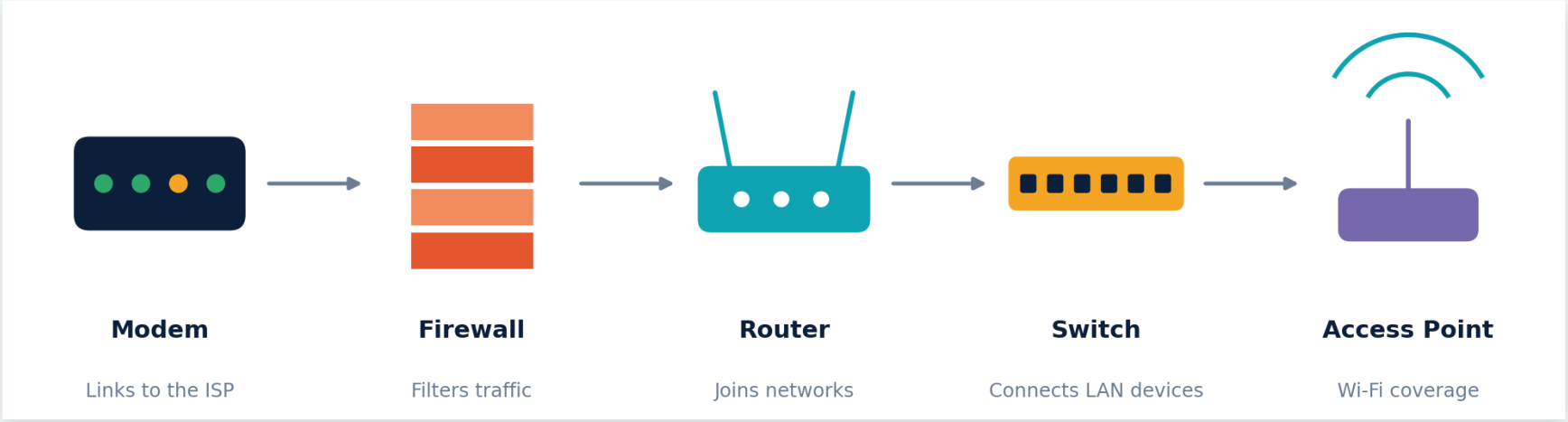
Beginnings research links between a few universities

Common language TCP/IP joined different networks

Everyone online the Web and Wi-Fi reached homes

Today billions of connected devices

Network Components



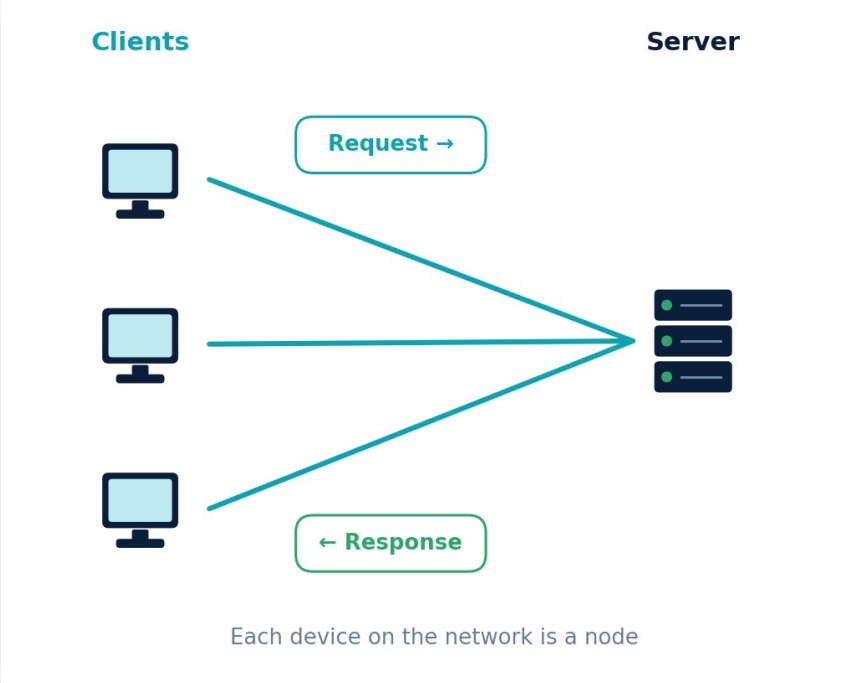
NIC gives each device its connection

Media cables or radio carry the signal

Switch and router move data inside and between networks

Modem, AP, firewall ISP link, Wi-Fi, protection

Clients, Servers, Nodes



Node any device on the network

Client asks for a service

Server provides web, files or mail

Many clients share one server

Types of Networks

PAN around one person, such as Bluetooth

LAN home, office or school

MAN a city or campus

WAN countries; the Internet is the largest



LAN vs WAN

	LAN	WAN
Area	Room / building	City to worldwide
Speed	100 Mbps – 10 Gbps	Slower, varies
Owner	One organization	ISPs / many
Cost	Low	High
Example	Home Wi-Fi, lab	The Internet

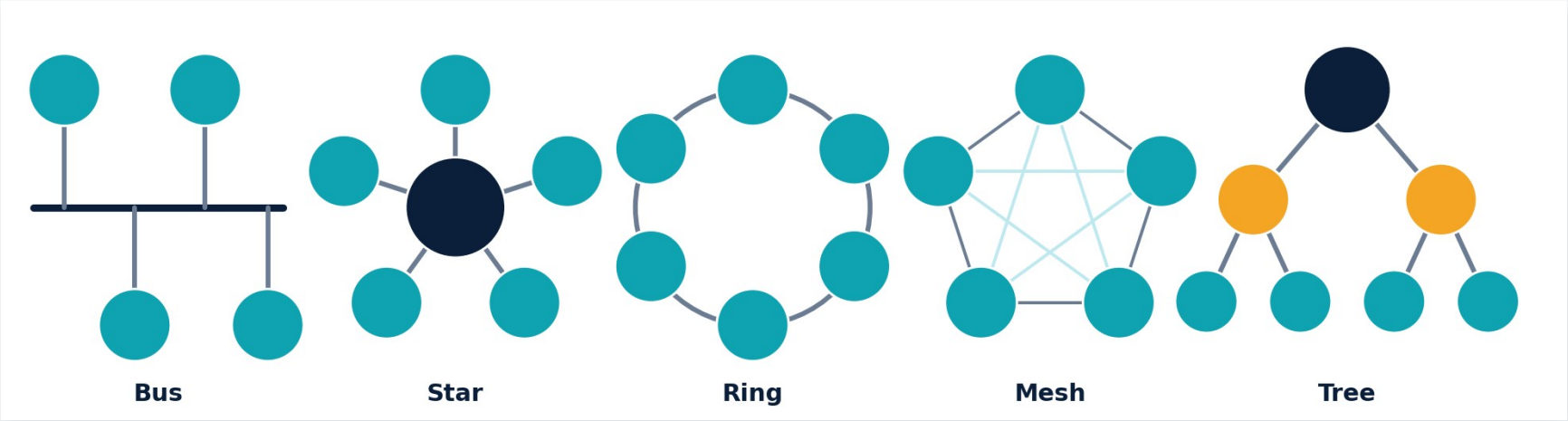
LAN small area, fast, privately owned

WAN wide area, slower, costlier, uses ISPs

Internet the biggest WAN

Joined WANs link many LANs together

Network Topologies



Physical how devices are laid out

Logical how data actually flows

Common types bus, star, ring, mesh, tree

Choice affects cost, speed and reliability

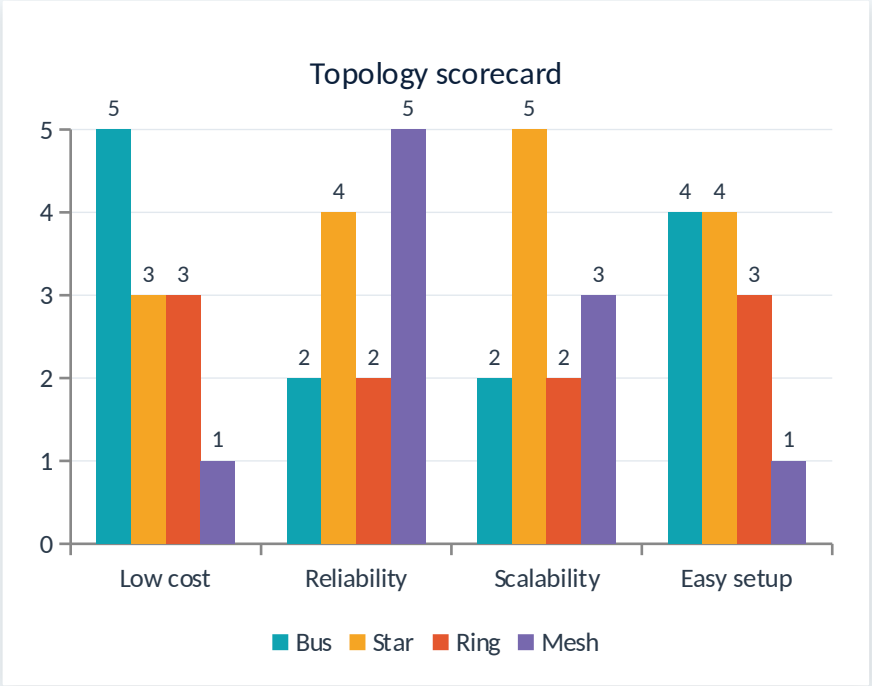
Comparing Topologies

Bus cheap, but one cable fault stops all

Star easy to manage; switch is the key point

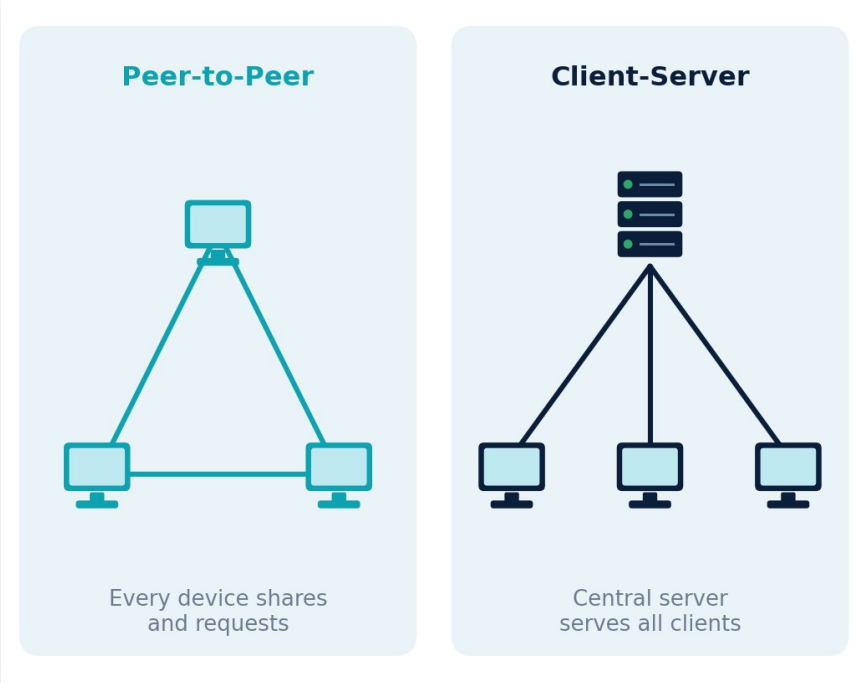
Ring orderly flow, but a break hurts

Mesh most reliable, most expensive



Scores are relative: 1 = low, 5 = high

Client-Server vs Peer-to-Peer



Client-server central control, easier security

Peer-to-peer every device is client and server

Client-server costs more, scales well

Peer-to-peer cheap, but hard to manage at scale

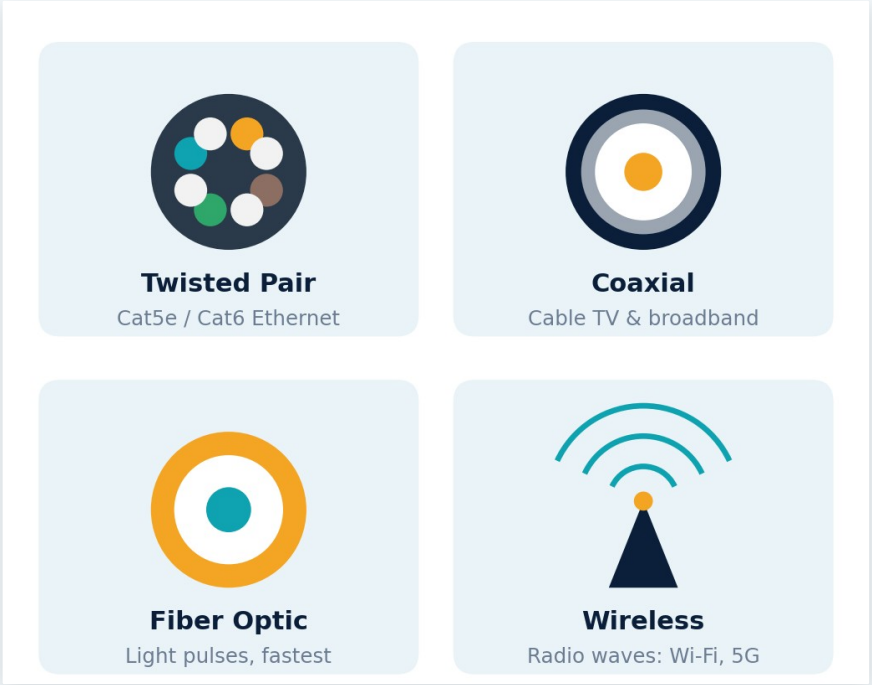
Transmission Media

Twisted pair common Ethernet cable

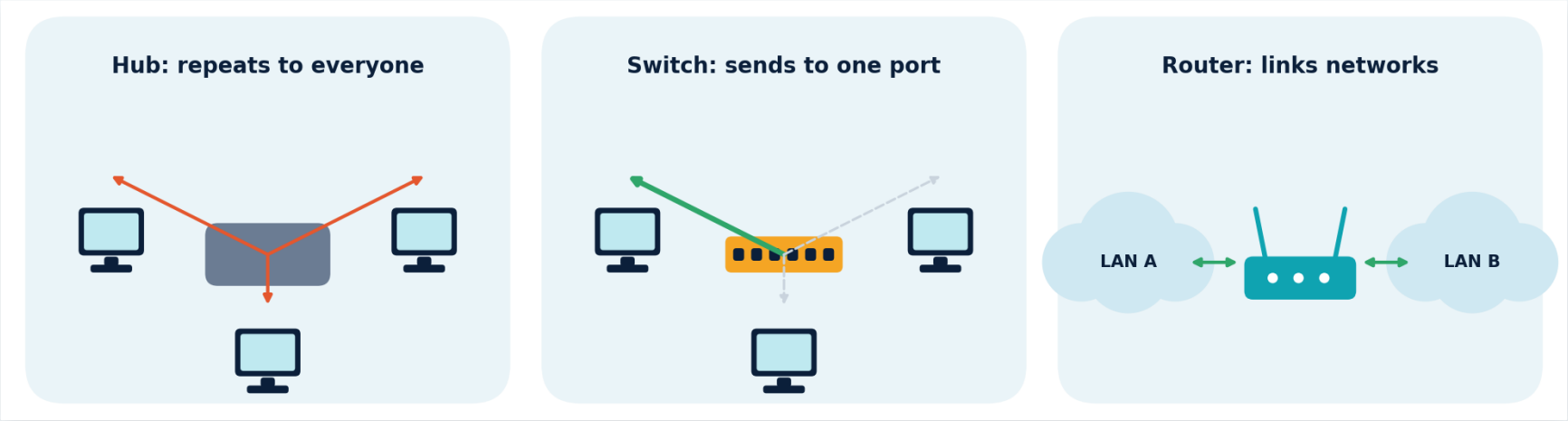
Coaxial cable TV and broadband

Fiber optic light signals, fastest, longest reach

Wireless radio waves: Wi-Fi, Bluetooth, 5G



Hub, Switch, Router



Hub repeats data to every port (obsolete)

Switch sends frames to one port using MAC

Router forwards packets between networks

Rule switch = inside a LAN, router = between LANs

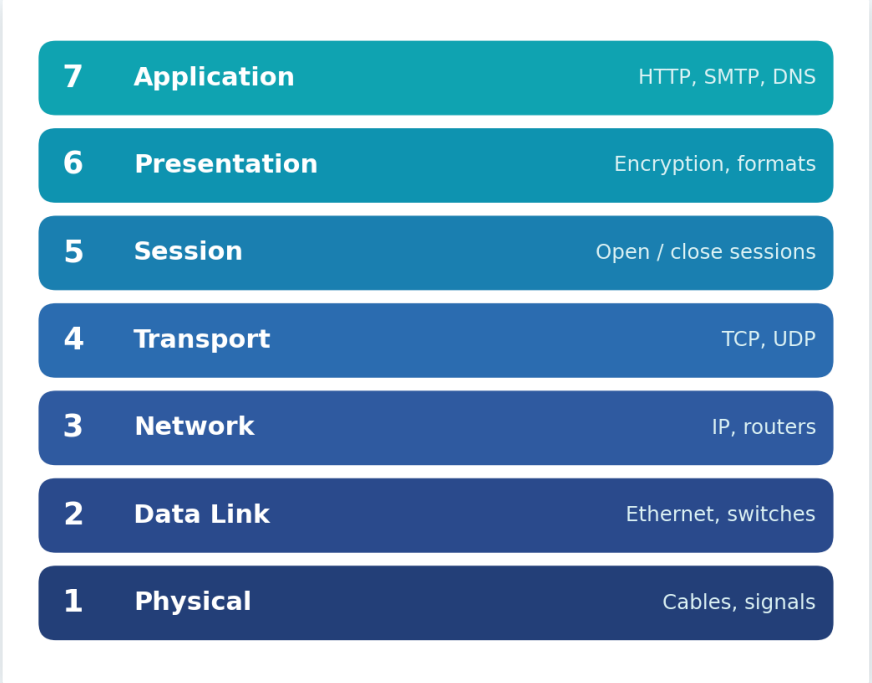
The OSI Model

Seven layers standardize how networks communicate

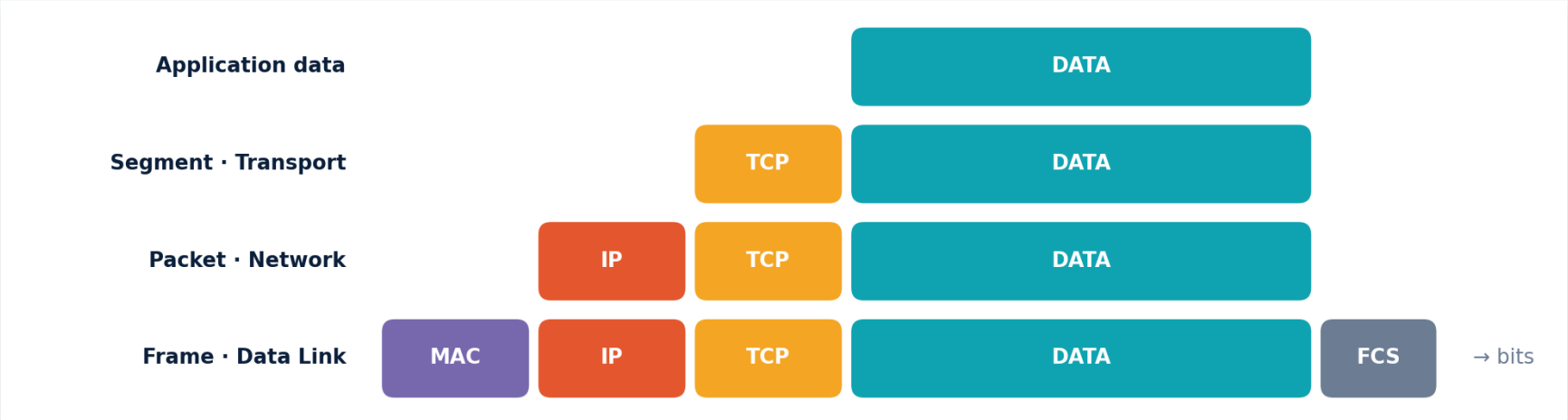
Top layers close to the user

Bottom layers move bits and frames

Mnemonic Please Do Not Throw Sausage Pizza Away



Data Encapsulation



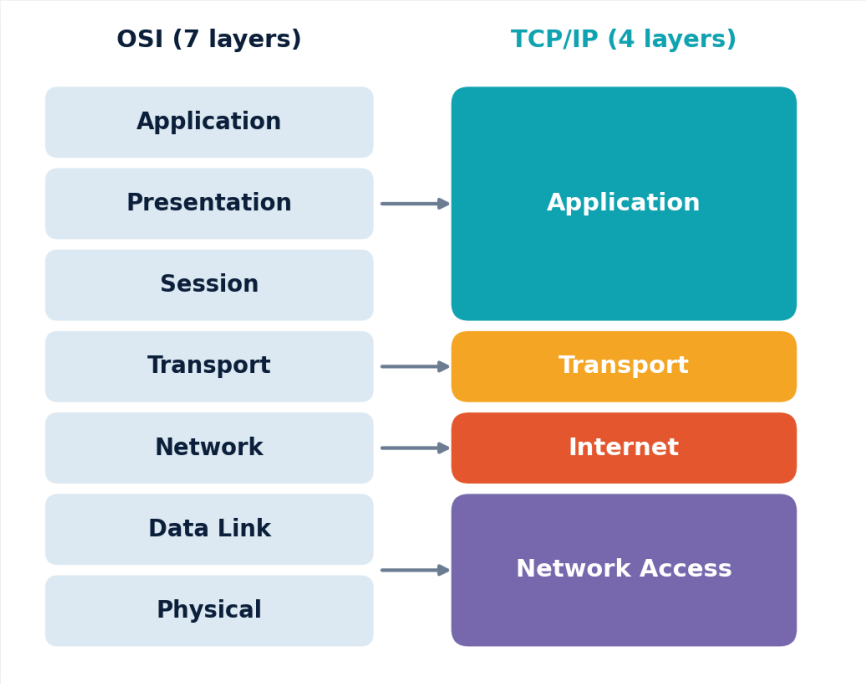
Going down each layer adds its own header

Receiver removes headers in reverse

Units data, segment, packet, frame, bits

Name encapsulation and decapsulation

The TCP/IP Model



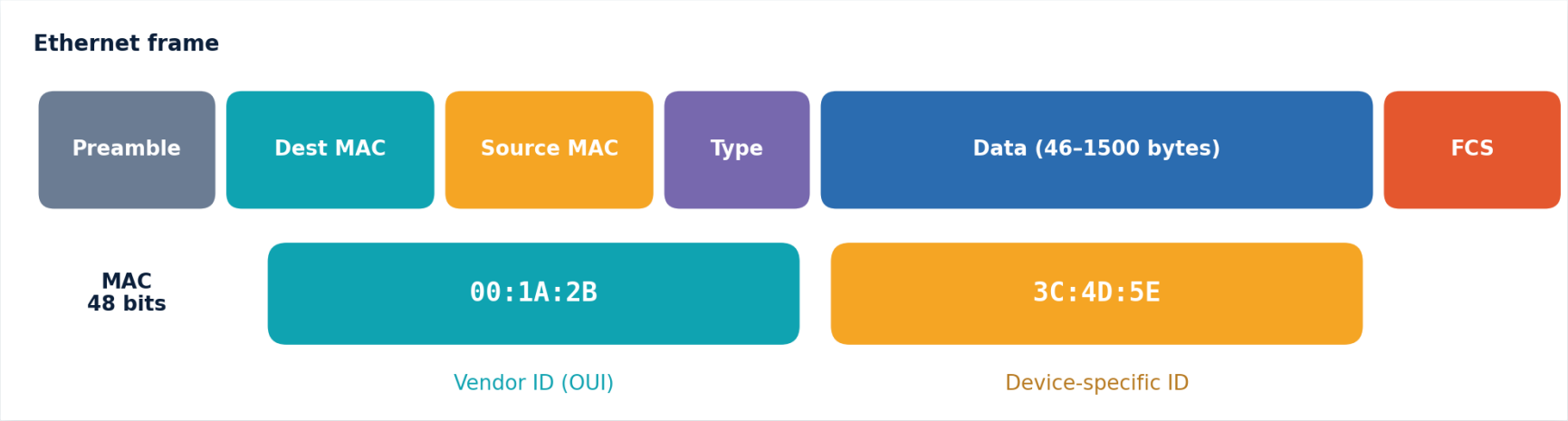
Four layers Application, Transport, Internet, Network Access

Practical the model the Internet runs on

Mapping combines several OSI layers

Same goal reliable end-to-end delivery

Ethernet and MAC Addresses



MAC 48-bit hardware address on every NIC

Format six hex pairs, such as 00:1A:2B:3C:4D:5E

Switches learn MACs to forward frames

Frame carries MACs, type, data and an error check

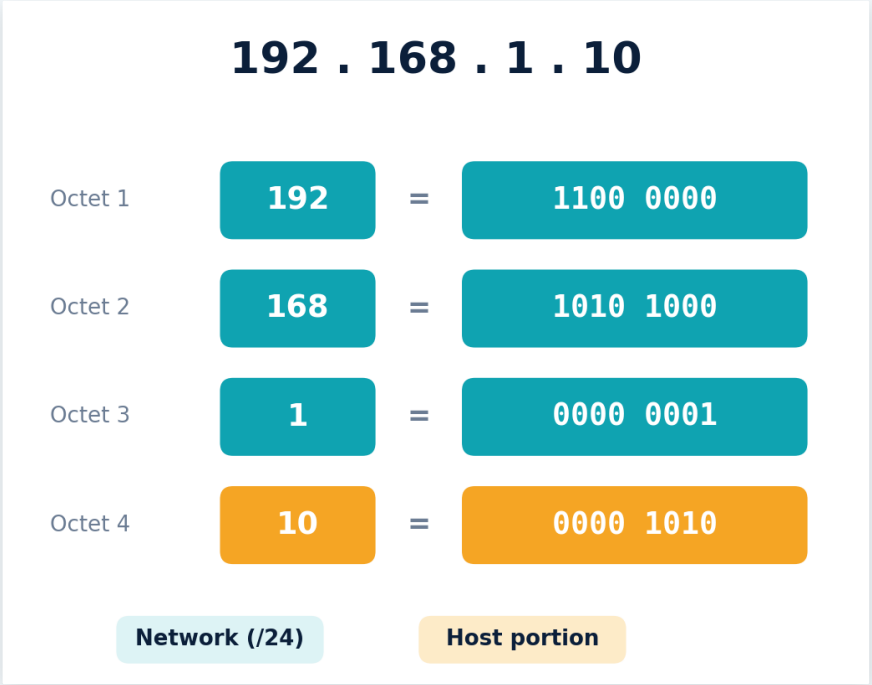
IP Addressing

IP address identifies a device on a network

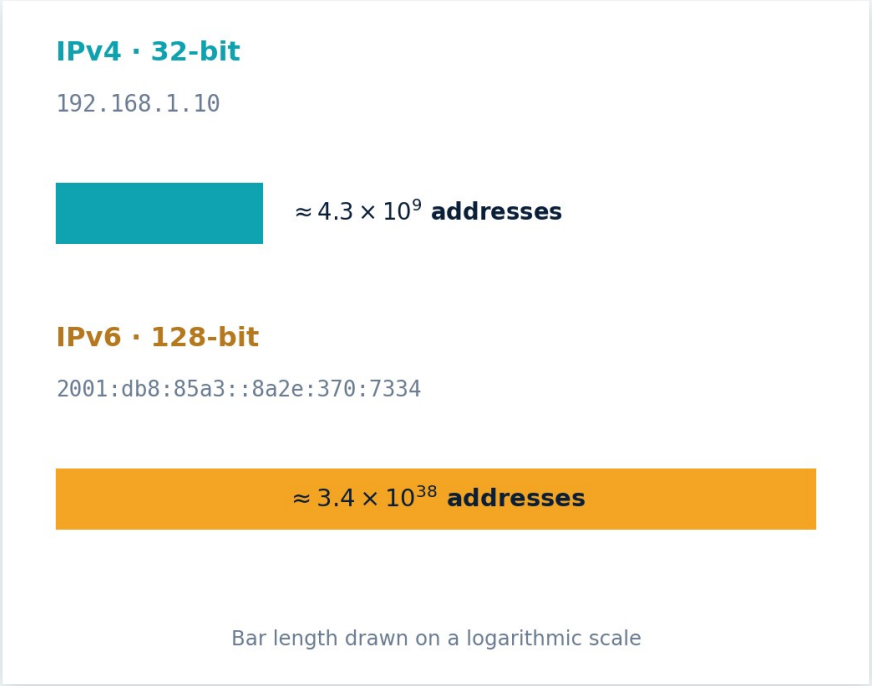
IPv4 32 bits, four octets

Two parts network portion and host portion

Private ranges 10.x, 172.16-31.x, 192.168.x



IPv4 vs IPv6



IPv4 32-bit, about 4.3 billion addresses

IPv6 128-bit, about 340 undecillion addresses

Why IPv6 IPv4 addresses ran short

Today both run together (dual stack)

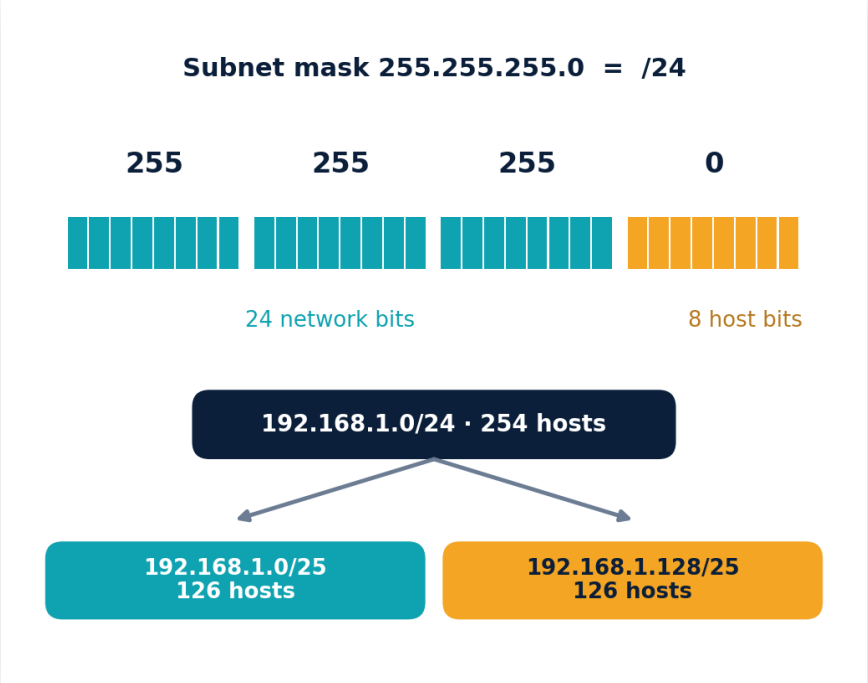
Subnetting

Idea split a network into smaller ones

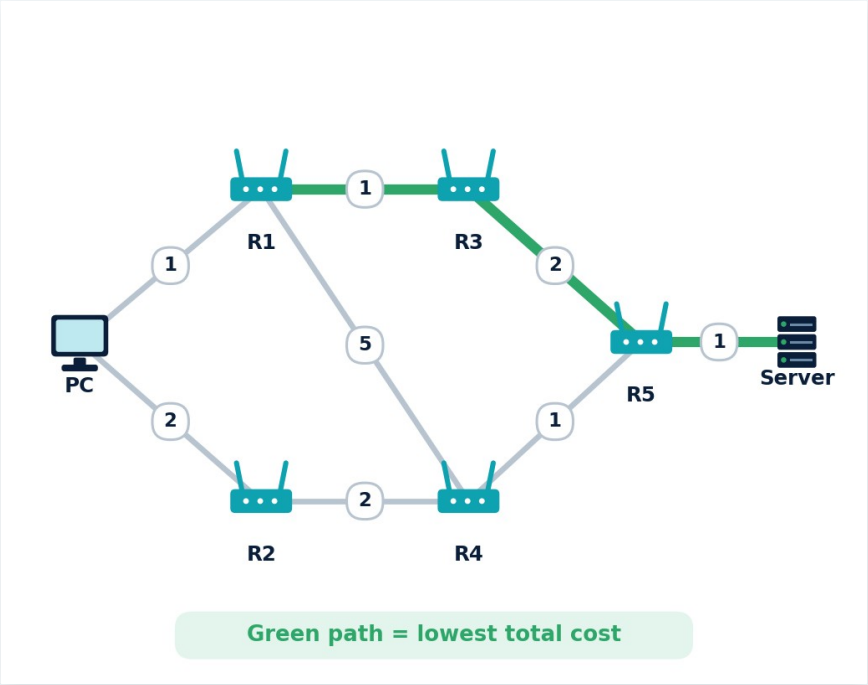
Mask 255.255.255.0 means /24

Hosts a /24 gives 254 usable addresses

Benefit less traffic, better security



Routing



Routers pick the best path from a routing table

Static set by hand; dynamic: RIP, OSPF, BGP

Gateway the exit door of your LAN

TTL stops packets looping forever

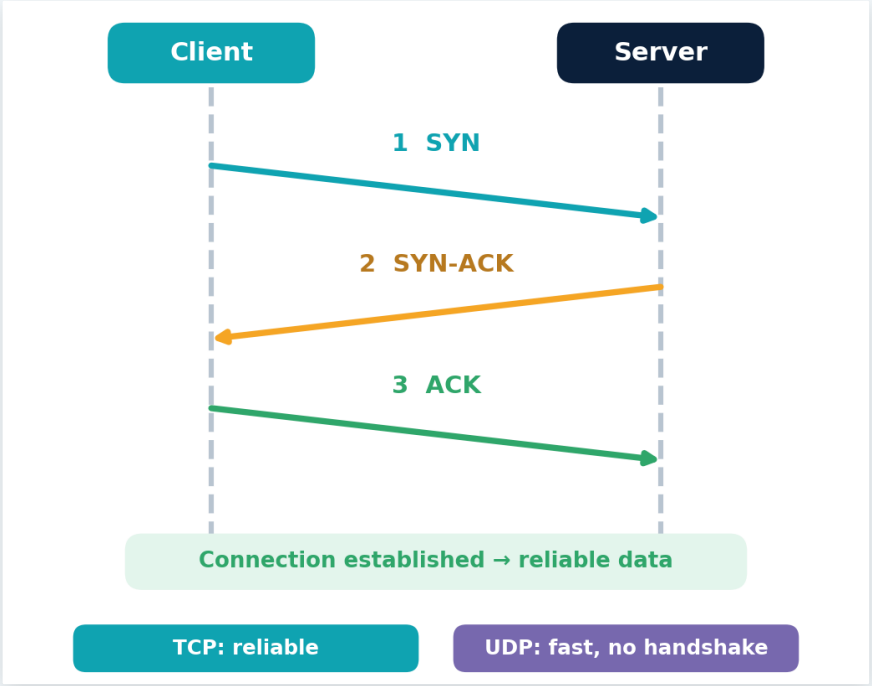
TCP vs UDP

TCP reliable and ordered; web, email

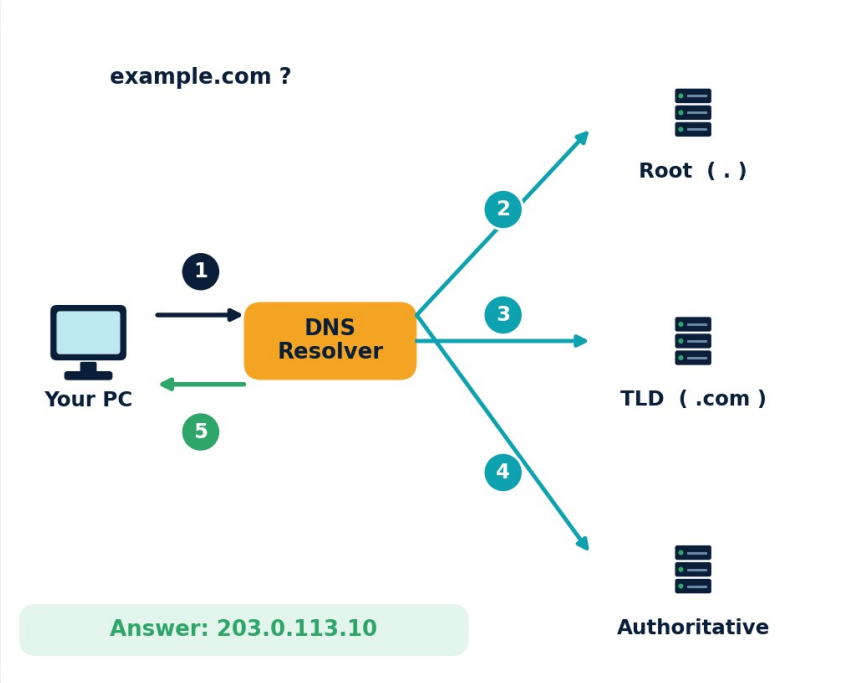
UDP fast, no guarantee; video, games, DNS

Handshake SYN, SYN-ACK, ACK

Pick accuracy (TCP) or speed (UDP)



DNS: The Internet Phonebook



Job turns names into IP addresses

Path resolver, root, TLD, authoritative server

Cache saves answers for faster lookups

Without it you would type numbers, not names

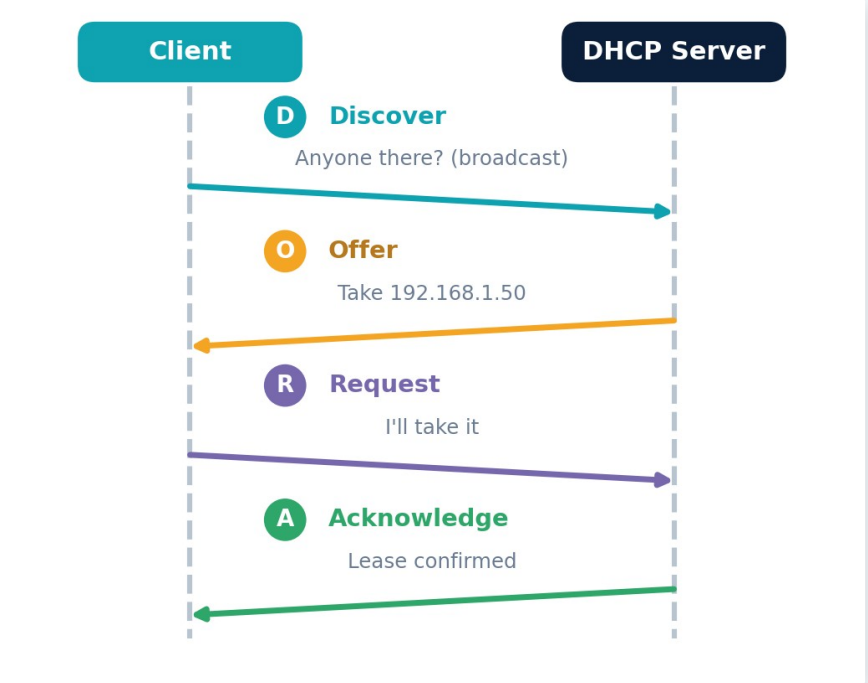
DHCP

Job hands out IP, mask, gateway and DNS

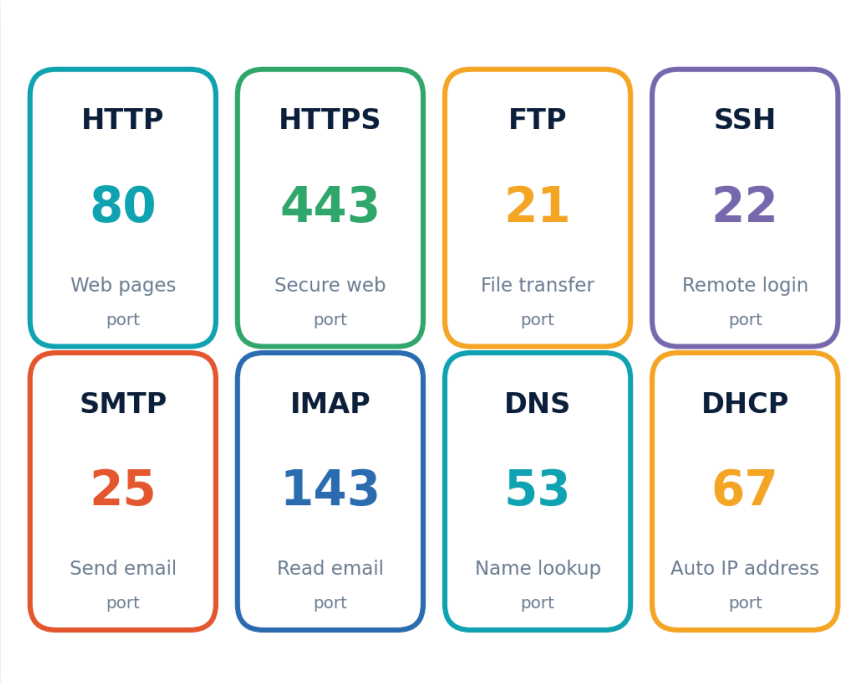
DORA Discover, Offer, Request, Acknowledge

Lease addresses are rented and renewed

Benefit no manual setup, fewer conflicts



Common Protocols and Ports



Port a numbered door to a service

Well-known ports 0 to 1023

Web HTTP 80, HTTPS 443

Others SSH 22, SMTP 25, DNS 53

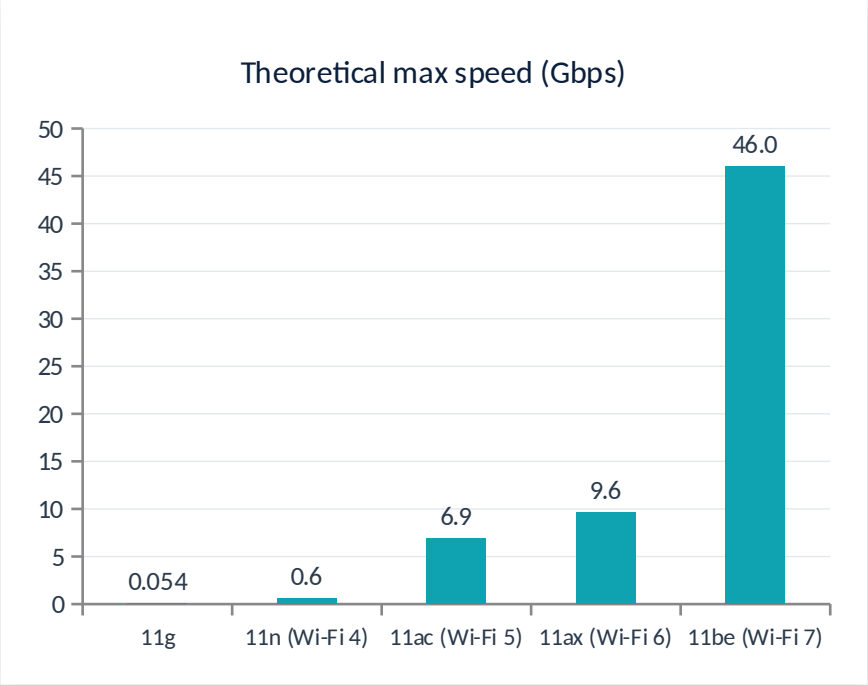
Wi-Fi Standards

2.4 GHz longer range, more interference

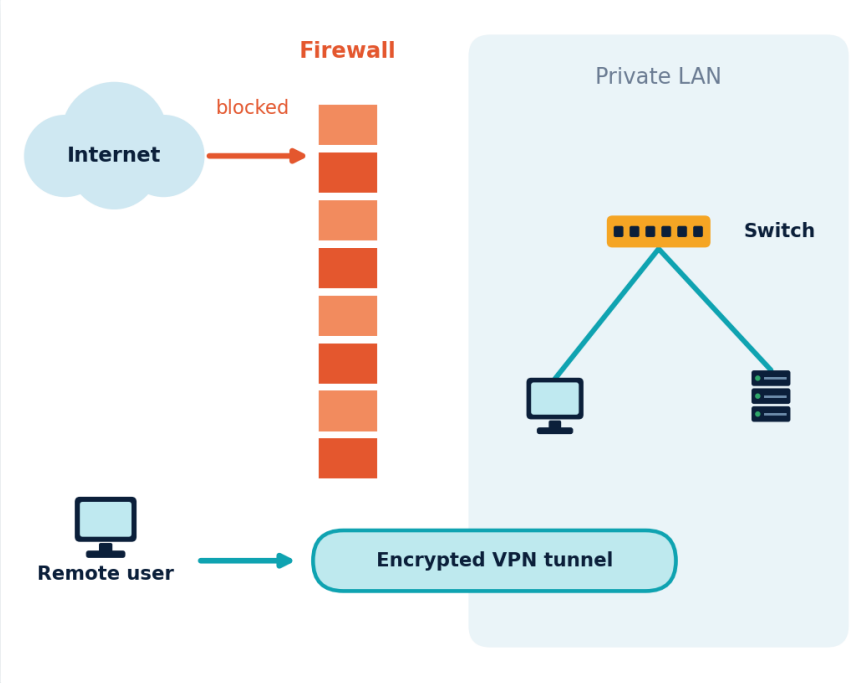
5 GHz faster, shorter range

6 GHz less crowded (Wi-Fi 6E and 7)

Security use WPA3, or WPA2 with a strong password



Network Security Basics



Firewall filters traffic by rules

VPN encrypts traffic over public networks

Encryption WPA3 and TLS protect data in transit

Access strong login and least privilege

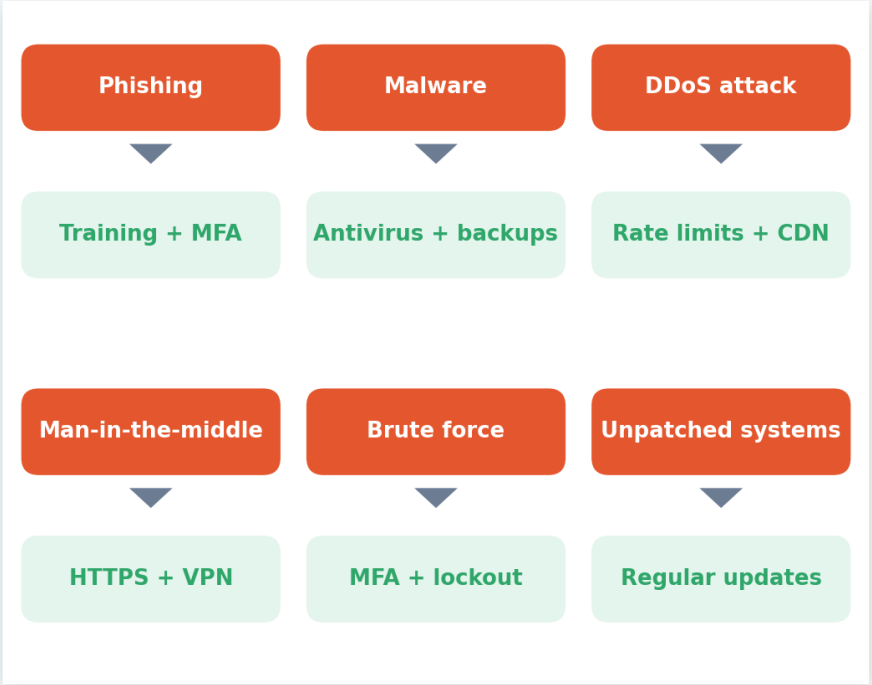
Common Threats and Defenses

Phishing, malware trick or infect users

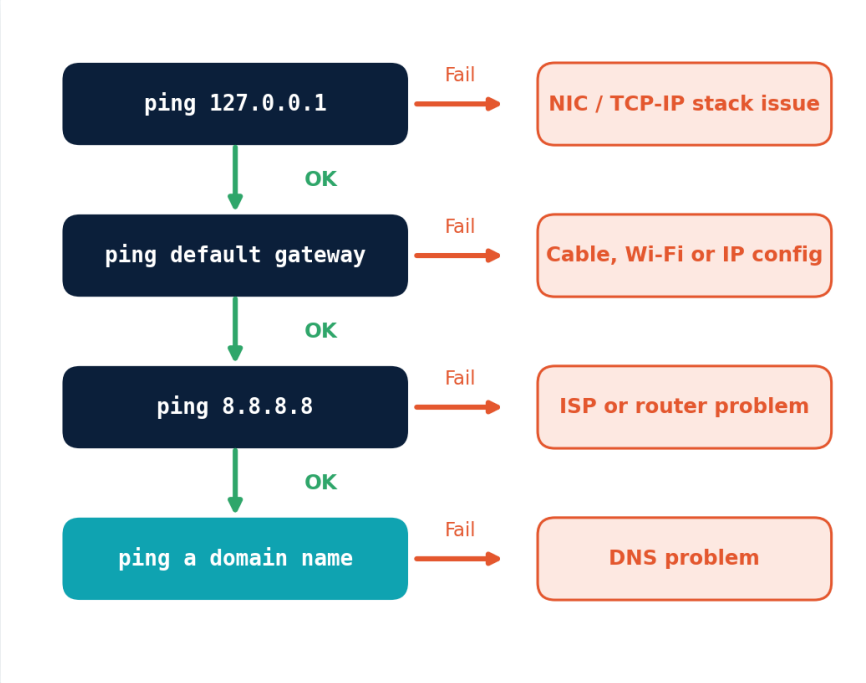
DDoS floods a service with traffic

Man-in-the-middle spies on traffic

Defense updates, MFA, backups, segmentation



Basic Troubleshooting



Method identify, isolate, fix, verify, document

ping tests reachability

tracert, ipconfig, nslookup path, settings, DNS

Work outward from your own device to the Internet

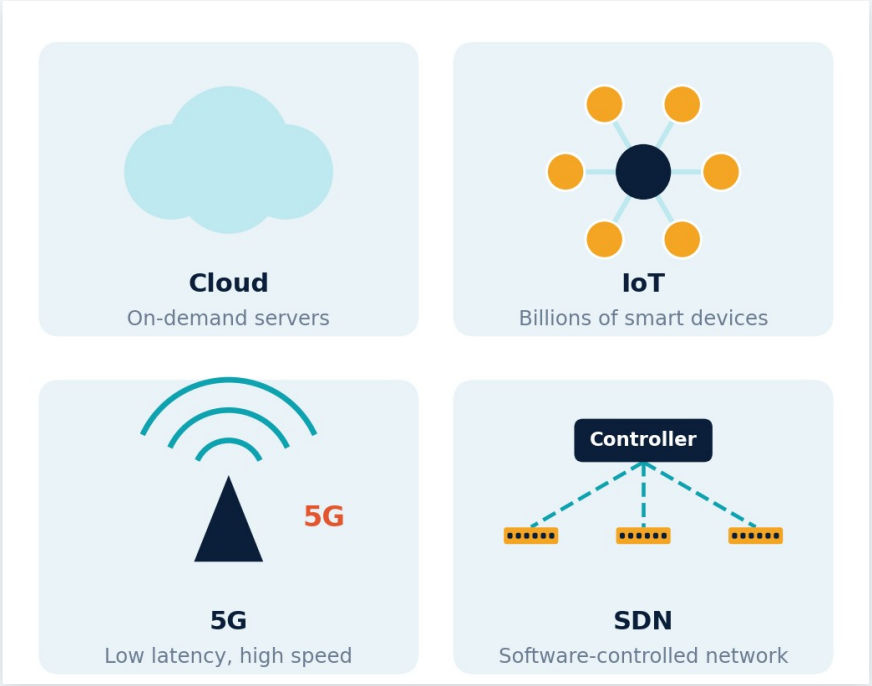
Modern Trends

Cloud rent servers and storage on demand

IoT billions of smart devices online

5G high speed, low latency wireless

SDN control the network with software



Key Takeaways

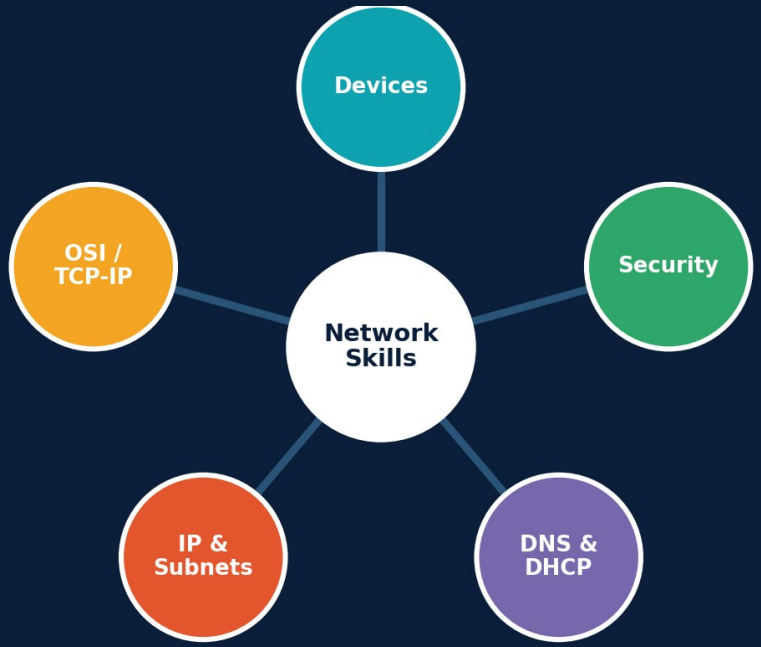
Devices and media build the network

OSI and TCP/IP give every layer a job

IP, subnets, routing find the way

DNS, DHCP, TCP make it usable

Security protects every layer



Thank You

Questions?

